Multiple Choice:

1. The common windows used in fiber optic communications are centered on wavelengths of

   A. 1300 nm, 1550 nm and 850 nm
   B. 850 nm, 1500 nm and 1300 nm
   C. 1350 nm, 1500 nm and 850 nm
   D. 800 nm, 1300 nm and 1550 nm

2. Bending losses:

   A. always result in breakage of the fiber
   B. can be caused by microbends and macrobends
   C. are used to detect the length of fiber on a drum
   D. are caused by the difference in the operating temperature of the core compared with the cladding on active fiber

3. Dispersion:

   A. causes the core to spread out and get wider as the pulse is transmitted along the fiber
   B. results in the wavelength of the light increasing along the fiber
   C. is the lengthening of light pulses as they travel down the fiber
   D. cannot occur with a laser light source

4. As the meridional ray is propagated along the optic fiber it:

   A. travels in a sort of spiral shape
   B. stays in the center of the fiber
   C. passes repeatedly through the center of the core
   D. is reflected off the inside surface of the primary buffer. This is called TIR

5. An active fiber detector:

   A. is used to prevent accidental exposure to invisible light
   B. is a communication system used during installation
   C. detects movement of fiber in security systems
   D. can be used to weigh objects

6. The primary buffer:

   A. reflects the light back into the core to allow propagation of light down the optic fiber
   B. must have a lower refractive index than the core
   C. is a plastic layer around the cladding to protect the optic fiber from mechanical damage
   D. must be airtight to prevent loss of the core vacuum

7. In a single-mode fiber cable, determine the maximum allowable diameter of the core for light frequency of 300 THz and numerical aperture of 0.35.

   A. 2.2 μm
   B. 2.5 μm
   C. 1.2 μm
   D. 1.5 μm

8. A transparent material along which we can transmit light is called

   A. a fiber optic
   B. a flashlight
   C. an optic fiber
   D. a xenon bulb

9. Absorption loss is caused by:

   A. insufficient stirring of the ingredients during manufacture
   B. changes in the density of the fiber due to uneven rates of cooling
   C. microscopic cracks in the cladding which allow leakage of the vacuum in the core
   D. impurities in the fiber

10. What is the definition of a bound ray?

    A. A ray that cannot move
    B. A ray that travels in the air
    C. A ray that is refracted out of the fiber
    D. A ray that propagates through the fiber by total internal reflection

11. What does a translucent substance do to light rays that fall on it?

    A. Reflects and absorbs them
    B. Refracts and absorbs them
    C. Transmits and diffuses them
    D. Transmits and reflects them

12. Medium 1 is made of glass, while medium 2 is made of ethyl alcohol. Their refractive indexes are 1.3 and 1.28 respectively. For an angle of incidence of 34 degrees, determine the angle of refraction.

    A. 36.41°
    B. 34.61°
    C. 31.64°
    D. 35.14

13. The window with the longest wavelength operates at a wavelength of approximately:

    A. 850 nm
    B. 1350 nm
    C. 1550 μm
    D. 1.55 μm

14. Besides the transmitter, the optical fiber, and the receiver, which of the following components may be found in a fiber optic link?

    A. Splices only
    B. Couplers only
    C. Connectors only
    D. Splices, couplers, and connectors

15. No material could have a refractive index of:

    A. 1.5
    B. 1.1
    C. 1.3
    D. 0.9

16. A cable containing both optic fiber and copper conductors is called:

    A. an armored cable
    B. a dehydrated cable
    C. tight-jacketed cable
    D. a hybrid cable

17. Intramodal dispersion:

    A. only occurs in multimode fiber
    B. is also called chromatic dispersion
    C. does not occur in multimode fiber
    D. could not occur in an all-plastic fiber

18. A simple fiber optic system would consist of

    A. a light source, an optic fiber and a photo-electric cell
    B. a laser, an optic fiber and an LED
    C. a copper coaxial cable, a laser and a photo-electric cell
    D. an LED, a cathode ray tube and a light source

19. If a ray of light approaches a material with a greater refractive index:

    A. the angle of incidence will be greater than the angle of refraction
    B. TIR will always occur
    C. the speed of the light will increase immediately as it crosses the boundary
    D. the angle of refraction will be greater than the angle of incidence

20. A white fiber with a black tracer is most likely to be fiber number:

    A. 18 C. 16
    B. 14 D. 12

21. An optical fiber and its cladding have refractive indexes of 1.55 and 1.32 respectively. Determine the acceptance angle.

    A. 54.34° C. 53.44°
    B. 35.33° D. 67.23°

22. If the wavelength of the transmitted light were to be decreased, the number of modes would:

    A. increase
    B. decrease
    C. remain the same
    D. halve in a graded index fiber

23. It is not true that

    A. endoscopes use coherent bundles of fibers
    B. silica glass is used because of its clarity
    C. a photocell converts light into electric current
    D. plastic fiber is normally used for long distance communications

24. The refractive index of a GI fiber:

    A. is at its highest value at the center of the core
    B. is usually higher in the cladding than in the core
    C. increases as we move away from the center of the core
    D. has a value of 4 instead of the 2 common in step index fibers

25. The speed of light in a transparent material:

    A. is always the same regardless of the material chosen
    B. is never greater than the speed of light in free space
    C. increases if the light enters a material with a higher refractive index
    D. is slowed down by a factor of a million within the first 60 meters

END